Statistics and Trends: Exploring Netflix Content Patterns

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Introduction:

In this report, I have analysed the Netflix Movies and TV Shows dataset to understand the different types of content available on Netflix. The main aim of this analysis is to explore how movies and TV shows are distributed, how content has changed over the years, and whether there are any relationships between numerical values in the dataset. The dataset includes 8,807 records with 12 columns, containing information such as content type, release year, rating, duration, country and genre. To understand the data more clearly, I have used three different visualisations: a bar chart, a line graph and a correlation heatmap. These graphs help to identify important patterns and trends in Netflix content. The results are based only on the available dataset and do not necessarily explain the reasons behind these patterns.

1. Distribution of Movies and TV Shows

From the bar chart, I found that Netflix has more movies than TV shows in the dataset. Out of 8,807 titles, 6,131 are movies, which make up 69.6% of the total, while 2,676 are TV shows, representing 30.4%. This means that movies are the most common type of content in the dataset. The number of movies is more than twice the number of TV shows, with a difference of 3,455 titles. Overall, the graph clearly shows that movies have a higher share than TV shows in the dataset used for this analysis.

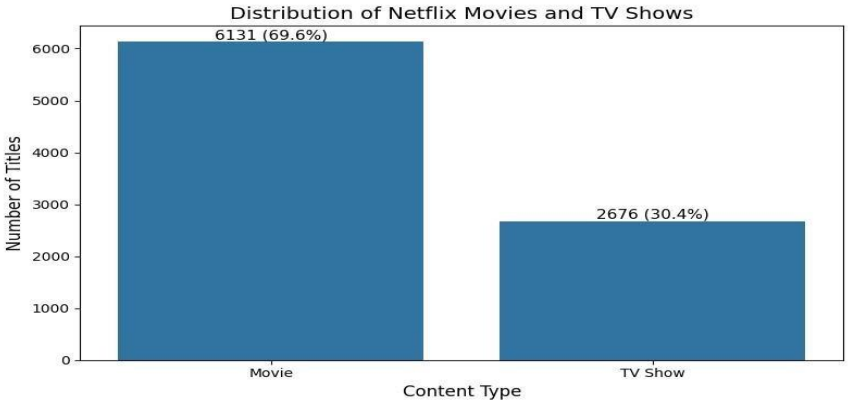


Figure 1. Number of Netflix titles grouped by content type

2. Netflix Titles by Release Year

The line graph shows how many Netflix movies and TV shows were released in different years. From the graph, I noticed that the highest number of titles was recorded in 2018, with a total of 1,147 titles. The number of titles gradually increased in the years before 2018 and started to decrease afterwards. This shows that a large number of titles in the dataset were released around this period. However, the decrease in the later years does not necessarily mean that Netflix produced less content. It may simply reflect the data available in the dataset used for this assignment.

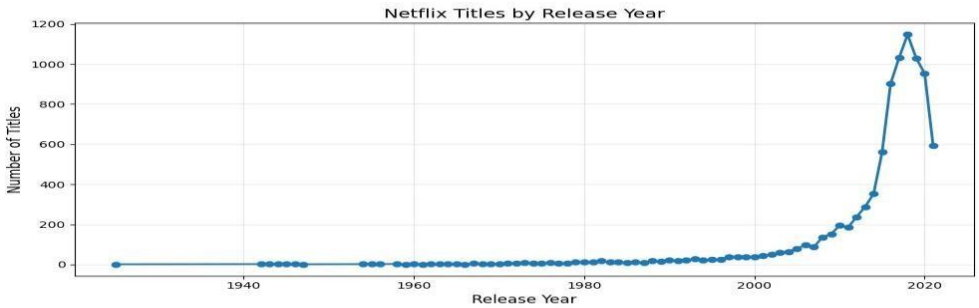


Figure 2. Number of Netflix titles by release year.

3. Correlation Heatmap

The correlation heatmap helps me understand how the numerical values in the Netflix dataset are related to each other. In this analysis, I have compared four variables: release year, title duration, the year the title was added to Netflix and the month it was added. The correlation values range from -1 to +1. Values closer to zero indicate that there is little or no linear relationship between two variables.

From the heatmap, I observed that release year and duration have the strongest relationship, with a correlation value of -0.249. This shows a weak negative relationship between them. The relationship between release year and the year a title was added is slightly positive, with a value of 0.112. I also found a weak negative relationship between the year and month a title was added, with a value of -0.160. The other variables have correlation values close to zero, which means there is very little linear relationship between them. Overall, the heatmap shows that most of the numerical variables in this dataset have weak correlations.

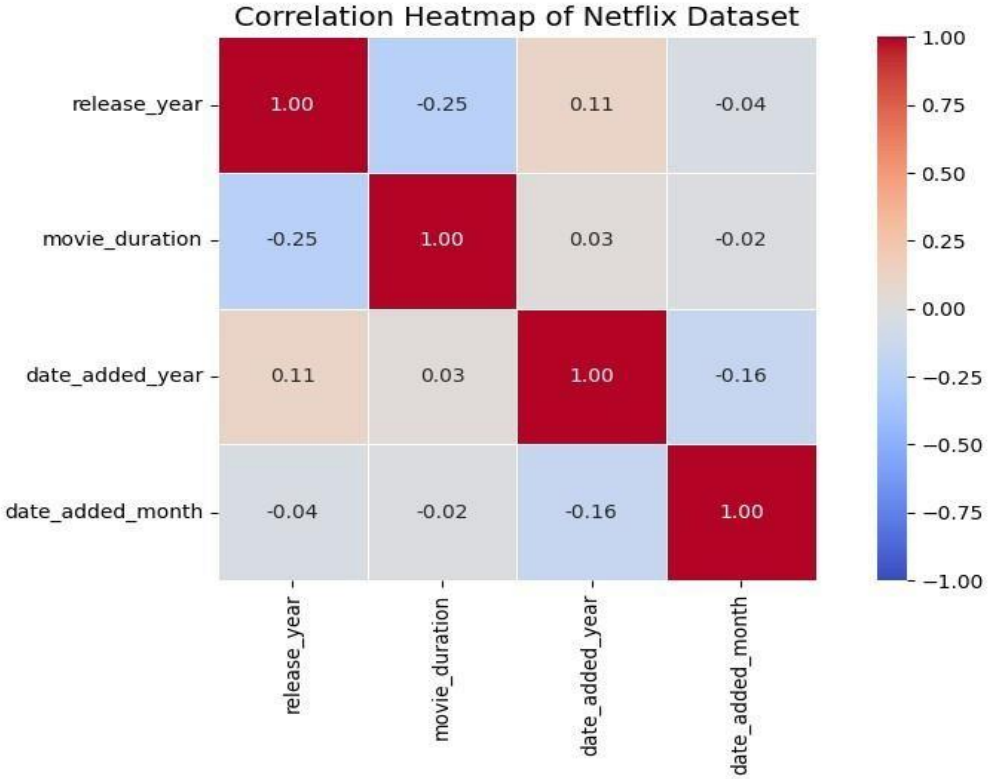


Figure 3. Pearson correlation heatmap for the four numerical variables.

Conclusion

After completing this analysis, I was able to identify a few important patterns in the Netflix dataset. Firstly, I found that movies make up the largest part of the dataset, accounting for 69.6%, while TV shows represent 30.4%. Secondly, 2018 had the highest number of released titles, with 1,147 records. Finally, the correlation heatmap showed that most of the numerical variables have only weak relationships with each other. The three graphs helped me understand the types of content available, the release-year patterns and the connections between different numerical values.

Overall, this analysis gave me a better understanding of the Netflix dataset. However, the results are based on the available data, and the duration column contains different types of measurements, which should be considered when interpreting the findings.